An Analytic Review of News Articles Covering the Israel–U.S. War with Iran

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Abstract. The Israel–U.S. war with Iran has become a focal point of the world since the beginning of March 2026. This paper focuses on sentiment analysis and machine learning using sentiment scores to capture what news articles are highlighting in key areas. New publications from four publishers are collected over 27 days. These articles are analyzed based on keywords to arrive at two conclusions: sentiment data alone can be used to predict one of the four publishers, and trends can be observed in sentiment over time. These objectives are met by retrieving news articles based on a first set of keywords. A second set of keywords is then used to identify sentences within each news article. These sentences are analyzed using sentiment analysis and averaged for each article across the keywords in the second group. Predictive analysis is conducted using inputs from these groups to determine the publisher. The results of the predictive analysis lend credibility to the average sentiment for the keywords. These sentiment trends are then used to assess overall tendencies of the publishers during the 27-day period of the war.

1 Introduction

This project will focus on the news domain covering the Israel–U.S. war with Iran. The data source will consist of news articles about the war from NBC, BBC, Fox News, and Al Jazeera. Transparency in the news is critical. The goal of this project is to determine how sentiment in Israel–U.S. war with Iran news articles has changed—or not changed—between March 14, 2026 and April 10, 2026 when a ceasefire was agreed to. Another objective is to examine whether these news outlets shift their tone and, if so, which ones do. An additional point of analysis is to determine whether there is sufficient signal in the sentiment of the data to predict which news outlet published a given article.

2 Methodology

2.1 Data Collection

The data source for this analysis is news articles from four international news outlets. Each outlet's RSS feeds and archive URLs from NBC [9] [8], BBC [4] [3], Fox News [6] [5], and Al Jazeera [2] [1] are queried using keywords pertaining to

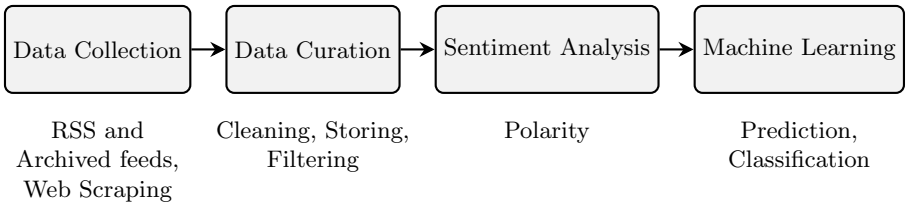

Fig. 1. Data Pipeline

the Israel–U.S. war with Iran. The keywords used are: Iran, Middle East, Israel, airstrike, airstrikes, missile, missiles, strike, strikes, and Hormuz.

The data format is text-based with associated metadata Table 1. The RSS feeds and archive URLs provide HTTP links to filtered news articles. Web scraping techniques are applied to retrieve the title, author, publication date, and full article text. The HTML was parsed using BeautifulSoup and CSS selectors. Newspaper3k was used for content extraction.

The specific output of the data collection process is a .csv file containing the features shown in Table 1. The primary features evaluated are title, author, publisher, publication date, and full article text. Additional features are included to assist in data curation. These include the article URL, source type, and keyword trigger. Headline length and article length are also recorded during the web scraping process and included in the .csv output.

Key -	Title Type	HL Pub Date	WC Coll Date	Author Link	Pub. Source	Ctry Keyword
2047	Three more Iran football team members change minds over asylum	62	604	Greenall - Atkinson	BBC	UK
	Public Broadcaster	3/14/26	3/15/26	bbc.com/...	RSS	iran
2060	Trump eases Russian oil sanctions as Iran war sends prices spiking	66	226	Alexander Smith	NBC News	US
	Broadcast Network	3/13/26	3/15/26	nbcnews.com/...	RSS	iran

Table 1. Example records from the collected news dataset. [7] [10] HL = headline length, WC = article word count.

2.2 Data Curation

The following steps were employed to curate the dataset: normalized special characters, calculated missing values, normalized the author field, applied publisher-specific cleaning logic, performed pattern-based text processing, normalized URLs and removed duplicates, and truncated full text into 300 and 600 word text blocks. These steps were critical for this data set, as the incoming data was different between the publishers and normalization was needed to make uniform sentiment analysis possible.

Unicode normalization and encoding repair were used to remove non-alphabetic characters. Author strings varied significantly between publishers, so unique rules were created to normalize names and remove additional author biographical details such as titles. For BBC News, the author's name was often surrounded by a date and additional characters; these details were also normalized. Publisher-specific cleaning functions were required to handle differences in byline formats. Regex-based pattern matching was used to eliminate metadata artifacts, descriptive clauses, and templated text. The full article was truncated into the first 300 and 600 words using tokenization to preserve complete sentences and enable comparison between longer and shorter articles. Approximately twenty percent of records contained the same article under two different URLs. This occurred because an article could appear in the RSS feed on one day and later on the archive site. URL canonicalization was used to normalize links by removing tracking parameters, allowing duplicate articles to be identified and removed so that each article appears only once in the dataset.

The retrieved data did not include article length or title length for Fox News articles. These values were computed and added to the dataset. Articles with full-text lengths of fewer than 50 words were removed from the dataset. These were removed because the full text resembled titles or short snippets rather than complete news articles.

News articles are currently retrieved nightly and collected from March 14 - April 11. The final dataset contains 652 unique articles with over 20,000 unique sentences. These met the keyword requirements and were gathered over 27 days.

The essential data attributes used in the analysis include article publisher, published date, author, title, full text, the first 300 words (start text), the first 600 words (half text), keyword sentences, and calculated sentiment analysis values for full text, start text, half text, and keyword sentences. The keyword sentences use a different set of keywords than those used to filter articles. These sentences are counted and assigned a sentiment score for each occurrence. The keywords include: Iran, (Israel or Israeli), Hormuz, Supreme Leader, Tehran, Trump, and (U.S., United States, or US).

The dependent variables are the calculated sentiment analysis values for full text, start text, half text, and keyword sentences for each article. The independent variables are article publisher, published date, author, title, and full text.

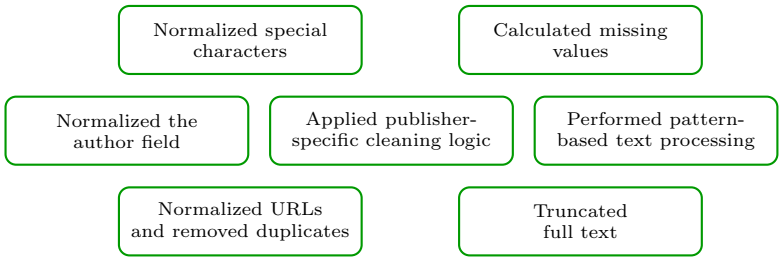

Fig. 2. Data Curation Processes

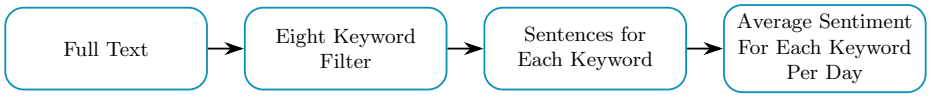

Fig. 3. Keyword-based sentiment processing pipeline.

2.3 Sentiment Analysis

One key aspect of the project was a feature engineering technique as a first step before the rest of the exploratory data analysis (EDA) portion could be completed. The full article text was broken into sentence groups based on keywords. After that function, sentiment analysis was applied using a Siebert transformer-based sentiment model with RoBERTa architecture Fig. 3. This model was selected since it is fine-tuned for sentiment classification. Next, another feature engineering technique was applied as follows: take the sentiment score and assign seven classifications Fig. 4. A Seaborn countplot function was visualized on these seven classifications. The results showed much higher counts for BBC and Al Jazeera, but that is a result of more articles being present in the dataset from them as compared to NBC and Fox News. The counts were normalized as a percentage of occurrences per publisher. The results showed negative sentiment dominating throughout the dataset.

Slight data cleaning was needed for the published date. If it was blank, the collection date was used. The retrieve article script is run daily, so most articles retrieved have a published date the same as the collection date. Also, articles without the keyword Iran in one of their sentences were filtered out of the dataset. This was to exclude Gaza, Ukraine, and North Korea articles. There were still some articles included in the dataset where the article selected was based on missile or Middle East. For example, if a Gaza article referenced the war with Iran, there could be a connection, so the article was included.

Other plots used were the Seaborn boxplot and time-based scatter plots with a mean reference line for each keyword and three text groups filtered by publisher. This data was more insightful as an overview of the mean, 25th and 75th quartiles, and range shown side by side. There are noticeable differences in the keyword sentiment numbers, which may be significant enough for a model to pick up on. Some results of visual inspection include: the sentences for the title

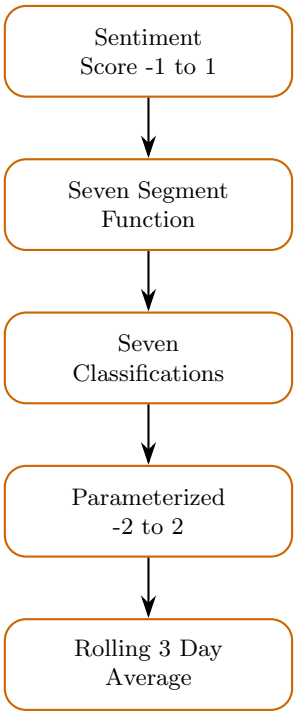

Fig. 4. Transformation of sentiment scores into parameterized classifications and smoothed averages.

keyword have the lowest value in terms of numerical value, but possibly analytical value as well since so many are close to -1. When the titles are positive, they typically come in as 1. Title does not look like a valuable input to the downstream models. The half-text parameter is good for distinguishing Fox News from NBC News, while full-text values narrow in on BBC and Al Jazeera. The start-text parameter has more variability than the other two text parameters. Based on these, all three parameters could have value in downstream models and should be included. The Israel parameter has higher sentiment values for Fox News and NBC over BBC and Al Jazeera and should be useful for decision-making. The Hormuz keyword sentences show low sentiment scores for NBC and Al Jazeera, while higher for BBC and Fox News. The Trump keyword sentences are highest for Fox News, then NBC, while the Iranian keyword is equal. Fox News has the highest sentiment score for the Iran keyword. The Trump keyword has good variety and should be helpful. There are not very many Supreme Leader sentences, so a smaller weight should be given to these sentiment scores. The United States sentences look helpful.

Regarding correlations of sentiment between keyword sentences and full text, Iran and Iranian were the most closely aligned with the full text at almost 0.5.

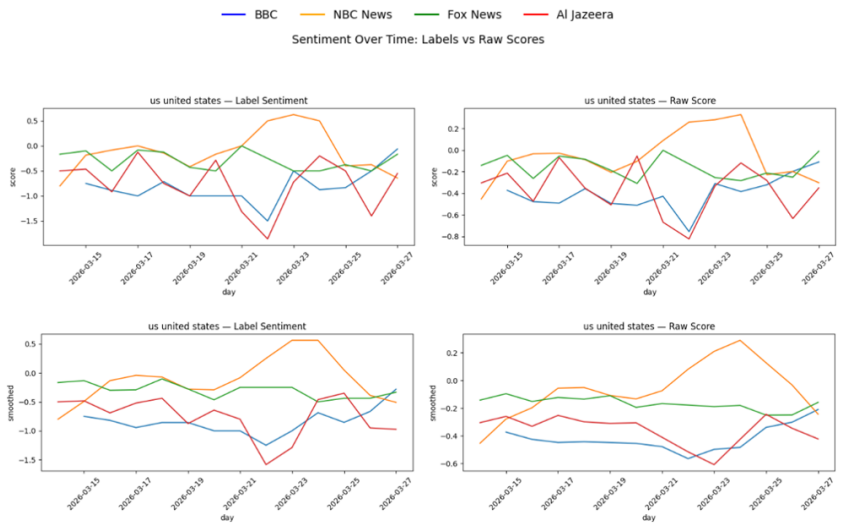

Fig. 5. The engineered feature on Left, Raw sentiment on Right

Feature engineering continued by taking the seven categories for sentiment and assigning them a score between 2 and -2 Fig. 4. This function was determined to give a larger range to the scores and give more weight to the highest or lowest sentiment scores. These were averaged, then smoothed with a three-day average and included as an engineered parameter. It was observed in the scatter plots that if there were only one or two sentences with a given keyword, the average for the day could be significantly different from other days with more keyword sentences. After considering the data in Fig. 5. and specific instances from the scatter plots, the smoothed engineered parameter was selected to be included in the model input.

There may be enough signal in the sentiment analysis to determine which publisher wrote the article because of the keyword sentences included. The full-text sentiment values were so similar between the publishers that it would be hard to distinguish between them based on that parameter alone. It remains to be seen whether the eight keywords selected and breaking the full text into two additional segments is helpful to the learning models. It appears that BBC and Al Jazeera will be the most difficult to distinguish. Determining NBC from Fox News will have the second hardest classification. Finally, distinguishing BBC and Al Jazeera from NBC and Fox News would be the easiest of the three.

2.4 Machine Learning Algorithms

A sentiment map is created to parameterize the sentiment string field. The features are defined for extraction from the sentiment file. The data is loaded

from the sentiment analysis file. The sentiment parameters are assigned to the DataFrame. The DataFrame includes only the selected features—published date, publisher, the raw sentiment score columns, and engineered sentiment columns. A small cleaning script ensures all fields are populated. Publisher names have spaces removed (e.g., Al Jazeera to AlJazeera). Time-based training is applied. The current day's articles are predicted using a Random Forest model. Daily accuracy is plotted over time. Accuracy and F1 score metrics are calculated for the current day. A confusion matrix is generated for the current day. Overall feature importance is also shown.

The Random Forest model was selected due to the non-linear sentiment values dominating the dataset. The model parameters were set to 300 estimators and a depth of 10. A key decision was choosing a class weight of "balanced," since Fox News has about 30 percent fewer articles per day than Al Jazeera and BBC. There were options to use distance features based on long-term and short-term profiles, but analysis showed that adding the engineered sentiment features outperformed these mathematical functions.

The purpose of this prediction model is to determine which publisher wrote an article from the current day. The training set explicitly excludes all data from today. It consists of records where the published date is less than the current day, where "day" is the most recent date in the dataset. The test set includes only records where the published date equals the current day. This training and test set approach is known as daily walk-forward live prediction. In this setup, the training set includes all data points prior to today. This broader dataset is maintained, as the full dataset currently contains about 500 articles, with approximately 35 new articles added for analysis each day.

The Daily Walk-Forward function trains on all prior days. It compares the current day's articles to the training set parameters and predicts the publisher. Metrics and predictions are stored. The code is designed to calculate an average accuracy for each day, including today. The input for any given prediction consists only of data from prior days.

The metrics from the final model were very satisfactory for sentiment-only input data. The accuracy for April 6, 2026, articles from the four publishers was 47 percent. F1 scores showed consistent ratios across all four publishers. The daily live prediction accuracy across dates was also satisfactory, if not exceeding expectations. The data suggests that the model improves as more training days are added, compared to the beginning when only a few days were available for training.

The URL for the Jupyter Notebook containing the model is: `https://github.com/hrawp/capstone_project/blob/main/notebooks/analysis_articles_Time_FeaturePlusBasic.ipynb`

A basic Random Forest model was selected to run using only sentiment from the full article and showed 29 percent accuracy. Accuracy increased to 37 percent when using 8 keyword sentiment scores and 4 text scores. However, when the 8 keyword and 4 text scores were passed through the feature engineering function and included with the initial feature set, accuracy improved to 45 per-

cent, which is a strong performance for sentiment-only inputs. The success of this third prediction set was based on creating the 8 keywords and, secondly, feature engineering a sentiment string and then parameterizing it. This served as a baseline model.

```
Latest prediction day:
          day  n_articles  accuracy  macro_f1  weighted_f1
23  2026-04-06          19  0.473684  0.467647     0.471827
```

	precision	recall	f1-score	support
Al Jazeera	0.44	0.50	0.47	8
BBC	0.40	0.67	0.50	3
Fox News	0.67	0.40	0.50	5
NBC News	0.50	0.33	0.40	3
accuracy			0.47	19
macro avg	0.50	0.47	0.47	19
weighted avg	0.50	0.47	0.47	19

Fig. 6. Accuracy with F1 Scores April 6, 2026

Next, the second and third configurations above were applied to time-based Random Forest models using daily walk-forward live prediction. The results are shown in Fig 6 and Fig 7. An interesting observation from the simpler models was the role of the engineered sentiment parameters. The data from the raw keyword scores produced better results than the engineered parameters alone. However, the engineered parameters showed higher accuracy on different days compared to the keyword scores. When combined, they created a strong complement and boosted accuracy by 7 percentage points. Both feature sets were then incorporated into the time-based model, resulting in exceptional performance Fig. 8.

The Random Forest library includes a feature importance function. Overall, the raw score columns carried the most weight, but the engineered columns played a key role in the final predictions. Interestingly, the importance of sentiment features differed from the score parameters, even though they were derived from the same sentences Fig. 9.

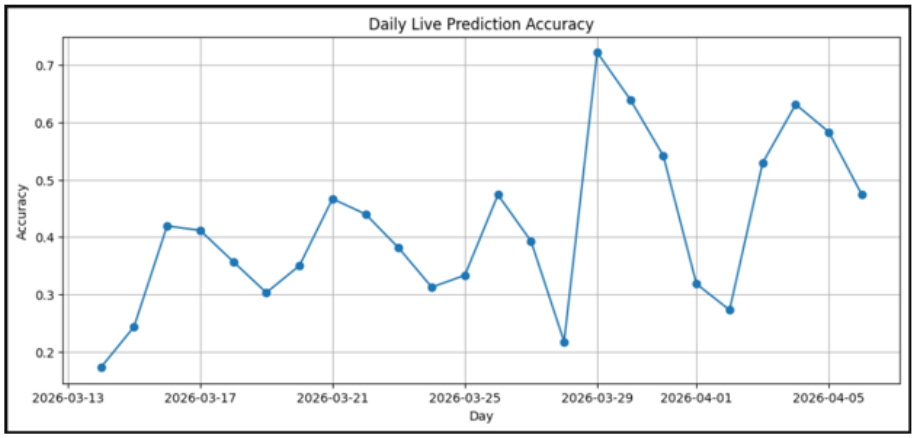

Fig. 7. Daily Live Prediction Accuracy

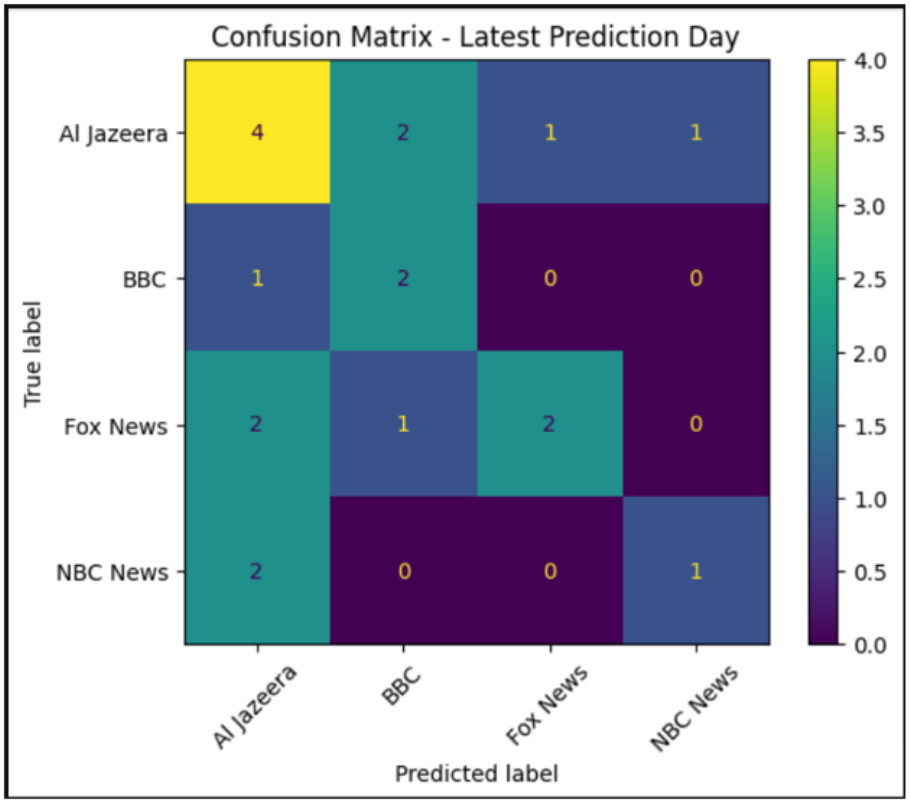

Fig. 8. Confusion Matrix April 6, 2026

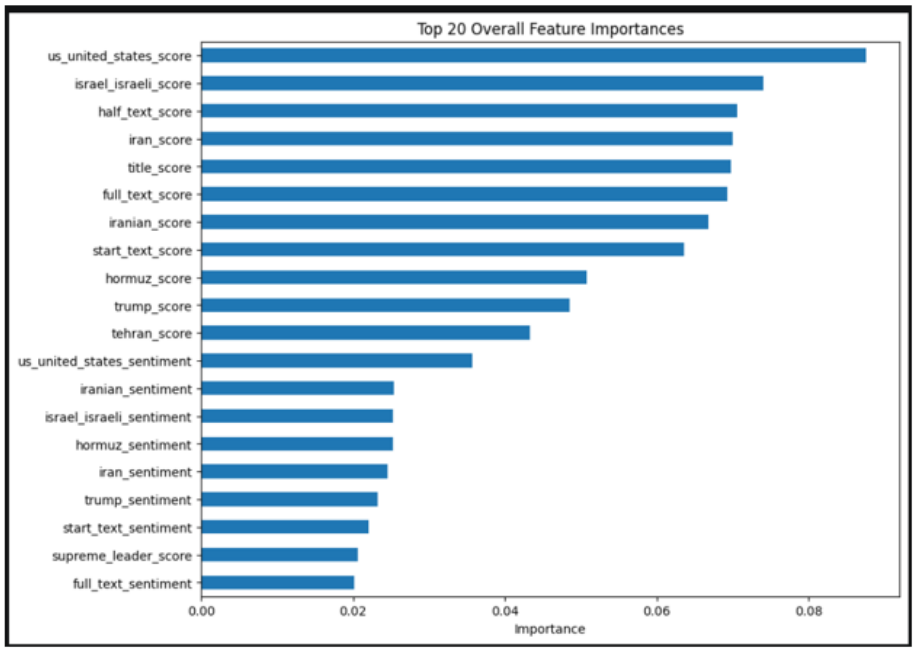

Fig. 9. Feature Importance

The source code for these methods can be found in hrawp Capstone Project GitHub repository at: `https://github.com/hrawp/capstone_project/tree/main`

3 Results and Analysis

There were two goals in the project. First, to see how sentiment shifts at the publishers over the time period (27 days). Second, to determine whether sentiment alone, as the only input, can identify which publisher created the news article with some level of accuracy.

To show how sentiment shifts over time, five charts are included. The first four focus on four keywords throughout the news articles: Iran, Israel, US, and Trump. These charts show the rolling 3-day average to reduce noise. The average per day is calculated from every sentence that contains a specific keyword for each article from the publisher. A full-text sentiment chart is also included.

For predicting the publisher, basic F1 scores, the confusion matrix, and daily accuracy scores provide a good representation of the quality of the sentiment-only approach. The detailed analysis of these scores was provided in the Machine Learning Algorithms section. These charts are attached again from four days later, at the ceasefire Fig. 15, Fig. 16, Fig. 17.

For the US keywords Fig. 10, sentiment was highest for the US-based publishers, Fox News and NBC, at around -0.2. BBC started with the lowest sentiment for the US, but it rose dramatically in the last 10 days. Al Jazeera reached the lowest sentiment for the US until the ceasefire negotiations progressed positively. These two publishers averaged around -0.4 for most of the period, with US sentiment reaching its highest level at the ceasefire.

For the US publishers, NBC had the higher sentiment, just under 0 early on, but its sentiment declined in the last 10 days. Meanwhile, Fox News maintained an average sentiment just under 0 during the last 10 days and even showed positive sentiment for three of those days.

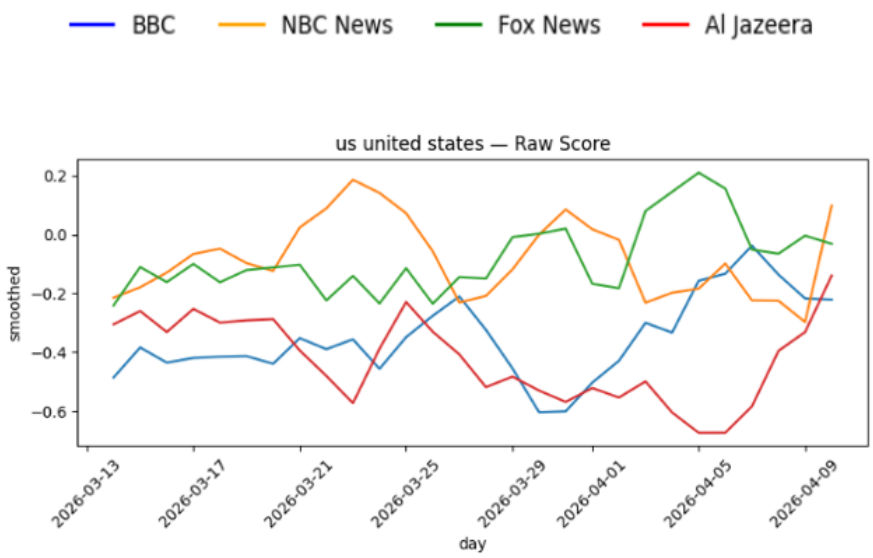

Fig. 10. US Keyword sentiment - 3 day rolling average

For the Israel keywords Fig. 11, sentiment was the lowest of all keywords, remaining below -0.4 for the majority of the time period. There was a spike between March 24 and April 2 for NBC and Fox News, but overall, these keywords had the lowest sentiment scores among those studied.

For the Iran keyword Fig. 12, BBC and Al Jazeera had the lowest sentiment values and closely mirrored each other until the end of the period. Fox News had the highest sentiment scores, remaining above -0.4 for most of the period. NBC and Fox News mirrored each other during the first half of the period, after which Fox News showed higher sentiment on average by nearly a 0.3 difference. Iran sentiment began increasing first for Fox News and then for BBC on April 1. NBC and Al Jazeera followed with a similar increase starting on April 6. The

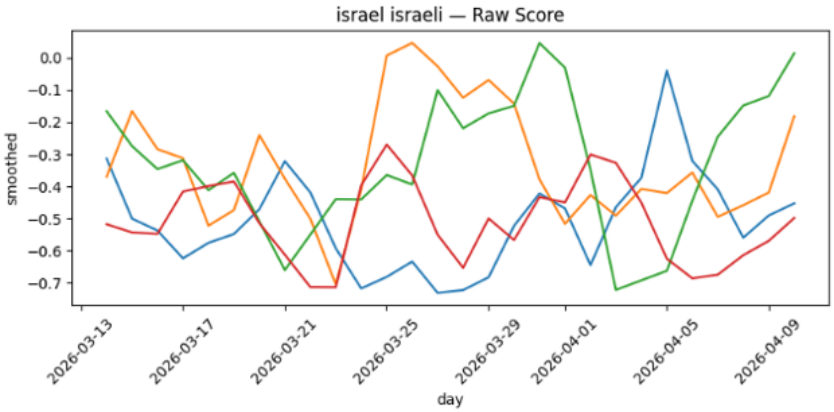

Fig. 11. Israel Keyword sentiment - 3 day rolling average

Iran keyword sentiment reached its highest value across all four publishers on April 10 during the ceasefire.

For the Trump keyword Fig. 13, sentiment at the beginning of the period averaged around 0 across the four publishers. It then decreased to approximately -0.2 for three weeks. By the end of the period, at the time of the ceasefire, sentiment for Trump returned to 0. Sentiment also changed in a similar manner across the publishers over shorter time spans.

For full-text average sentiment Fig. 14, values were above -0.2 for Fox News and NBC, while remaining below -0.3 for BBC and Al Jazeera. The 3-day average showed similarities between Fox News and NBC, as well as between BBC and Al Jazeera. Overall, changes over the time period for each publisher were minimal and remained within a narrow range.

For the prediction metrics, the accuracy levels were above 40 percent for eight consecutive days and on eleven of the last thirteen days Fig. 16. Accuracy was much better in the last half of the time period than in the first half. For the daily live prediction accuracy, only days prior to the test day were used for training. As more training days were added, accuracy improved over time.

Finding keywords and their associated sentences was key to this project. Trends can be observed, as shown in the data. This approach provided much more detail than using full-text sentiment alone. Since the publisher could be predicted more accurately with the use of keywords, this suggests there is a correlation between these sentiment values and adds credibility to the charts analyzed. There is meaningful signal and significance within these long article texts that can be leveraged for analysis.

The prediction model continued to deliver solid results in the last half of the time period, even when there were large shifts in sentiment across all publishers leading up to and during the ceasefire. This supports a strong conclusion that

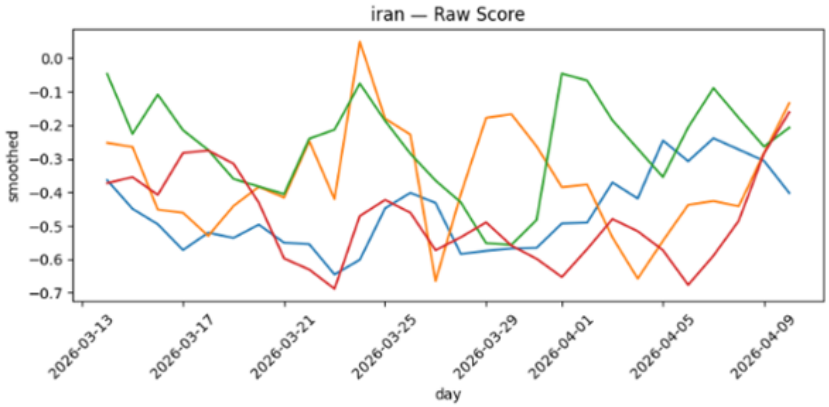

Fig. 12. Iran Keyword sentiment - 3 day rolling average

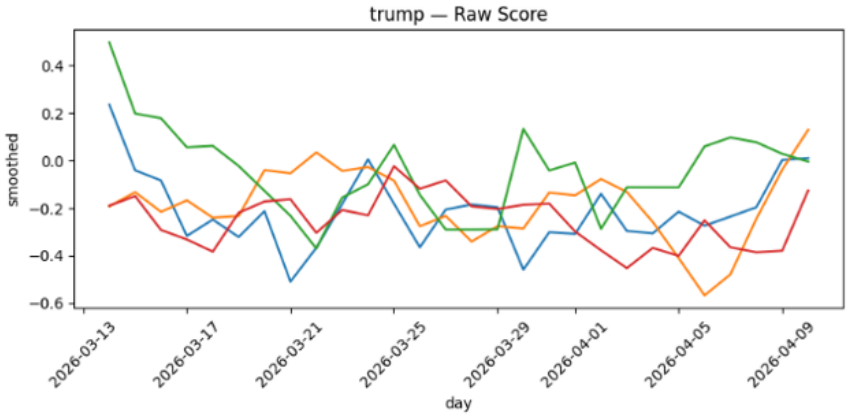

Fig. 13. Trump Keyword sentiment - 3 day rolling average

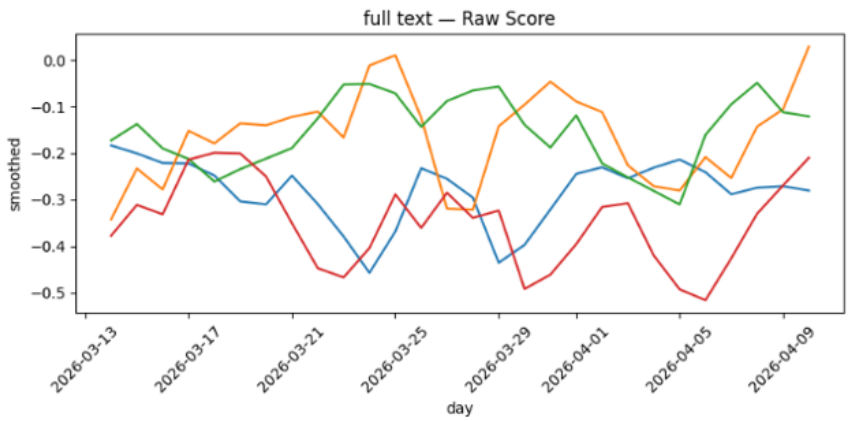

Fig. 14. Full Text sentiment - 3 day rolling average

Latest prediction day:

	day	n_articles	accuracy	macro_f1	weighted_f1
27	2026-04-10	20	0.5	0.505742	0.488908

	precision	recall	f1-score	support
Al Jazeera	0.50	0.44	0.47	9
BBC	0.33	0.25	0.29	4
Fox News	0.67	0.67	0.67	3
NBC News	0.50	0.75	0.60	4
accuracy			0.50	20
macro avg	0.50	0.53	0.51	20
weighted avg	0.49	0.50	0.49	20

Fig. 15. Accuracy with F1 Scores April 10, 2026

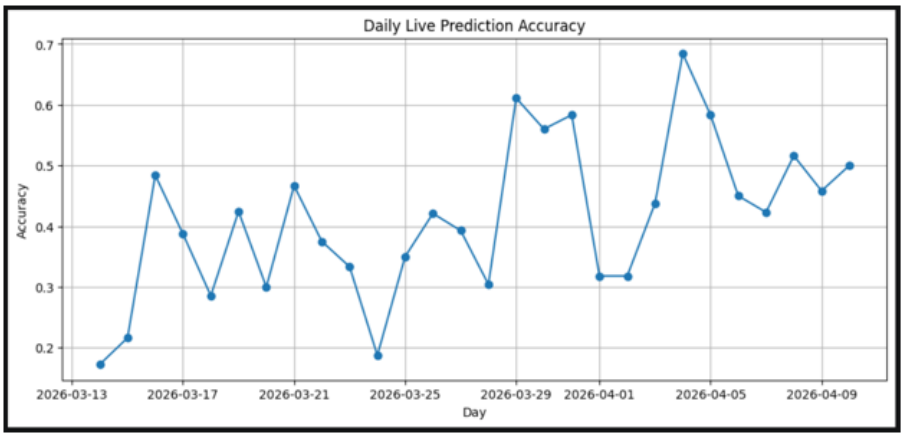

Fig. 16. Daily Live Prediction Accuracy

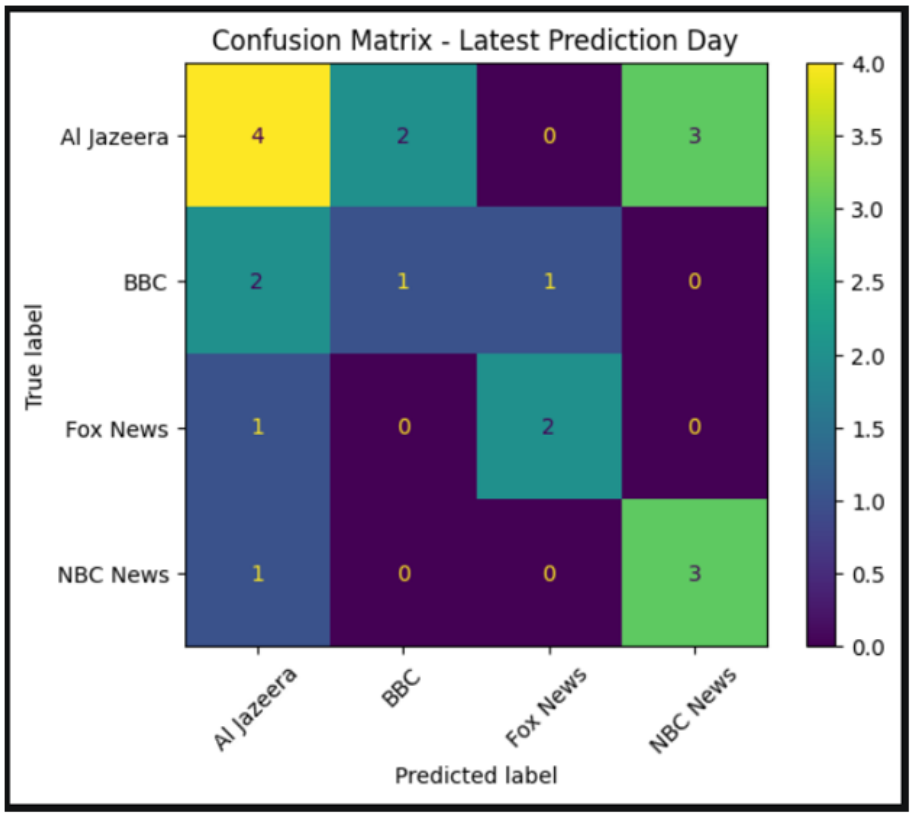

Fig. 17. Confusion Matrix April 10, 2026

the selected sentiment keyword analysis provides sufficient signal to determine the publisher.

4 Limitations

New articles were not available before March 14 for most publishers when the data collection script was fully implemented. The intent of the project was to examine how sentiment changed over the entire war, but the first 10 days of data were not able to be retrieved for analysis. A total of 27 days were reviewed, with a logical stopping point at the ceasefire. There were enough articles to complete the project; however, if the full date range had been analyzed, the final results may have differed. These differences could have been reflected in a stronger predictive model and more comprehensive sentiment analysis.

Another limitation was the use of only sentiment scores as inputs to the predictive model. More robust language models could be used to determine the publisher. However, the results using only sentiment scores were compelling and aligned with the intended scope of this project.

5 Conclusion

BBC and Al Jazeera consistently had lower sentiment scores across the keywords and full text. It did not significantly matter which keyword was chosen; they consistently had the lowest averages. These lower averages could indicate that these international publishers have a less favorable perception of the Israel-U.S. War with Iran than the US publishers.

Fox News had the highest sentiment values overall. This could be interpreted as more favorable toward the war than others. Alternatively, Fox News may have simply used fewer low-sentiment scoring words in their sentences.

The prediction model provides strong support for the average sentiment analysis over time for the keywords. Specifically, it indicates that the chosen sentiment keyword method is valid, and the conclusions above have a high degree of credibility.

Finally, the ceasefire had a significant impact on sentiment scores. Higher scores were observed across all four publishers when the agreement was made. This was seen in the main four keyword scores and full-text sentiment. It appears that all four publishers wrote more positively about moving towards a resolution to the conflict.

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